UNIVERSITY OF ENGINEERING AND TECHNOLOGY

TAXILA



Computer Engineering Department

Control System lab report #06

Submitted to:

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Submitted by:

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Task #1:

Construct a transfer function For R =4Ω,L=1H,andC=0.1F,construct the transfer function:□(□) = □_□(□)/□_□(□) = 1/(□□□² + □□□ + 1) Tasks: Use tf; display G, poles, DC gain, and plot the step response.

Code :

```
clc;
clear;
close all;

R = 4;
L = 1;
C = 0.1;

G = tf(1, [L*C R*C 1]);

disp('Transfer Function:');
G

disp('Poles:');
pole(G)

disp('DC Gain:');
dcgain(G)

disp('Damping and Natural Frequency:');
damp(G)

disp('Step Response Information:');
stepinfo(G)

figure;
step(G);
grid on;
title('Step Response of Series RLC Circuit');
xlabel('Time (seconds)');
ylabel('Capacitor Voltage (V)');

figure;
pzmap(G);
grid on;
title('Pole-Zero Map');
```

OutPUT:

```
Damping and Natural Frequency:

         Pole              Damping       Frequency      Time Constant
                                      (rad/seconds)      (seconds)

 -2.00e+00 + 2.45e+00i     6.32e-01       3.16e+00         5.00e-01
 -2.00e+00 - 2.45e+00i     6.32e-01       3.16e+00         5.00e-01
Step Response Information:

ans =

  struct with fields:

        RiseTime: 0.6124
    TransientTime: 1.8951
     SettlingTime: 1.8951
      SettlingMin: 0.9106
      SettlingMax: 1.0769
        Overshoot: 7.6893
       Undershoot: 0
             Peak: 1.0769
         PeakTime: 1.2894
```

```
Transfer Function:

G =

           1
  ------------------
  0.1 s^2 + 0.4 s + 1

Continuous-time transfer function.
Model Properties
Poles:

ans =

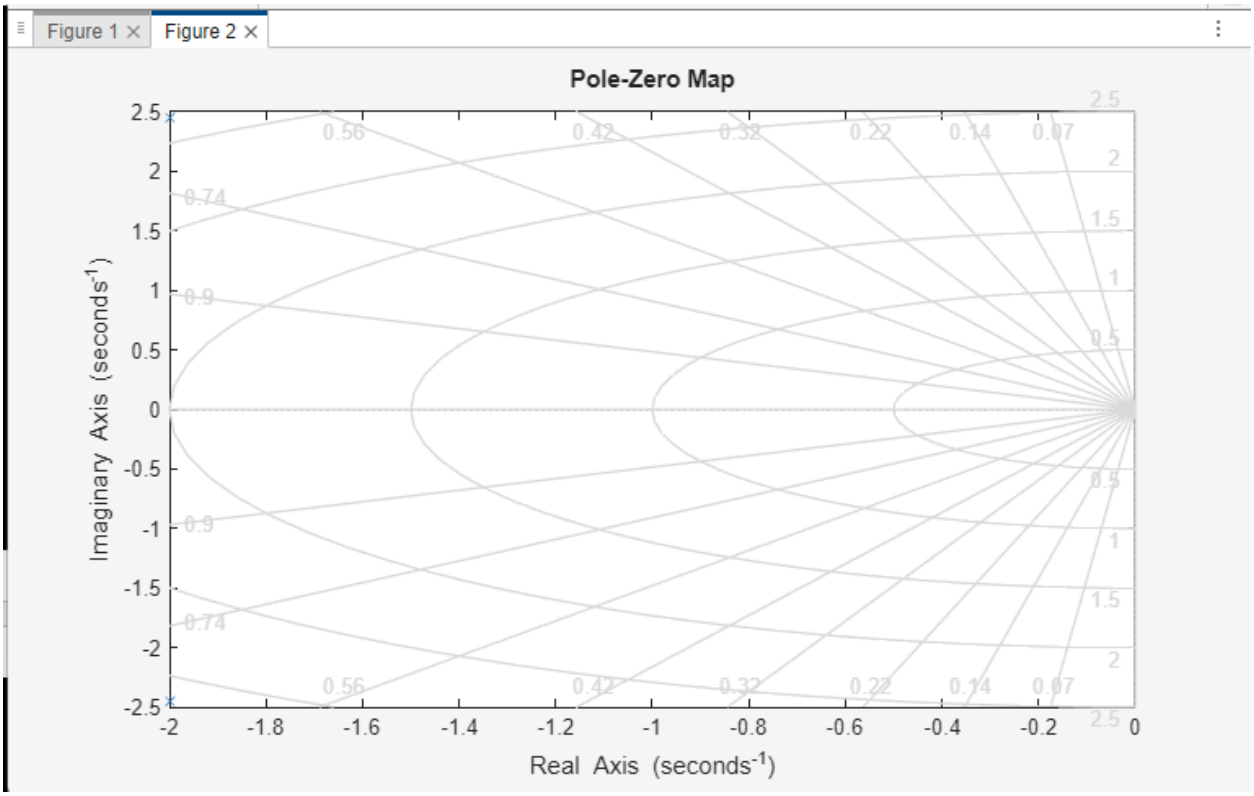
  -2.0000 + 2.4495i
  -2.0000 - 2.4495i

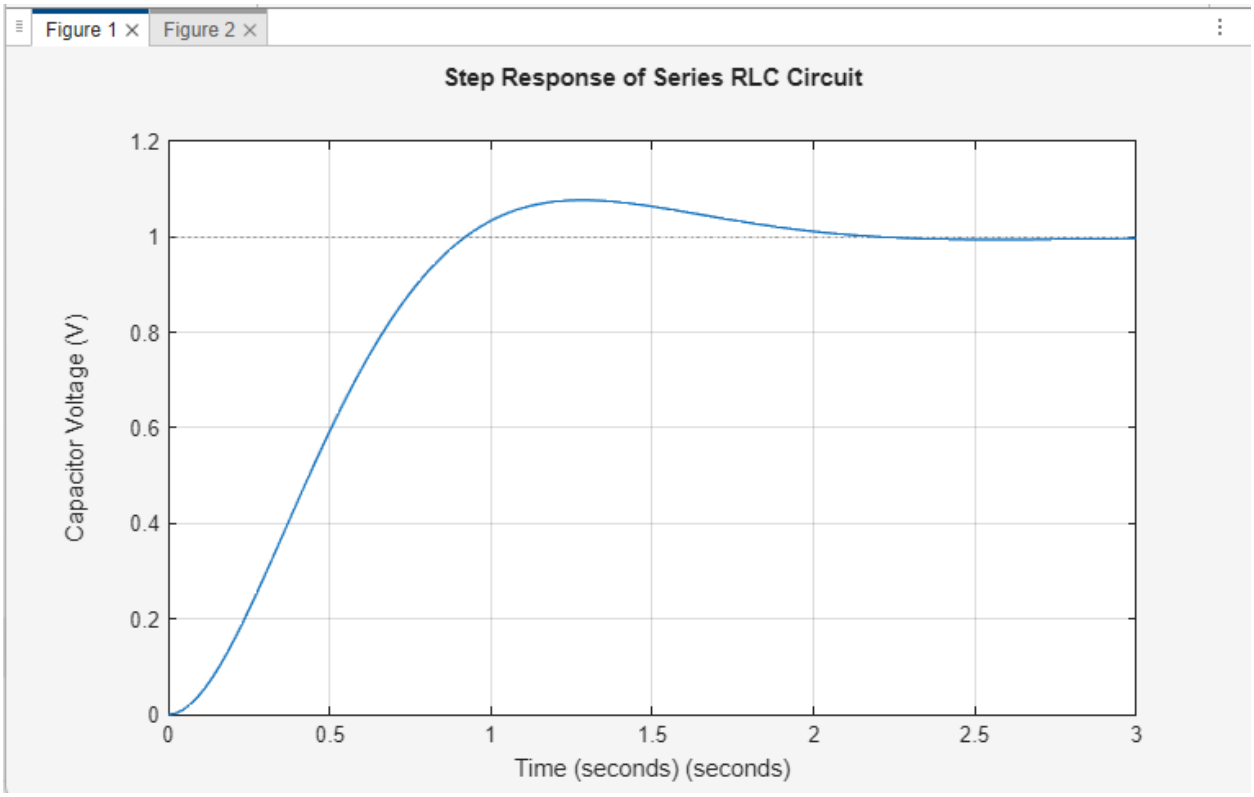
DC Gain:

ans =

     1
```

Figure:


Figure 1
Figure 2
Pole-Zero Map
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.56
0.42
0.32
0.22
0.14
0.07
0.74
0.9


Figure 1
Figure 2
Step Response of Series RLC Circuit
Capacitor Voltage (V)
Time (seconds) (seconds)

Task #2: dentify damping class For R =8Ω,L=0.5H,andC=0.02F, calculate:□_□ = 1/√(□□), □ = (□/2)√(□/□) Tasks: Verify using damp(G), then classify the system.

```
clc;
clear;
close all;

R = 8;
L = 0.5;
C = 0.02;

G = tf(1, [L*C R*C 1]);

disp('Transfer Function:');
G

disp('Natural Frequency and Damping Ratio:');
wn = 1/sqrt(L*C);
zeta = (R/2)*sqrt(C/L);

disp(wn);
disp(zeta);

disp('Poles:');
pole(G)

disp('Damping Verification:');
damp(G)
```

```
disp('DC Gain:');
dcgain(G)

disp('Step Response Information:');
stepinfo(G)

figure;
step(G);
grid on;
title('Underdamped RLC Step Response');
xlabel('Time (seconds)');
ylabel('Capacitor Voltage (V)');
```

Output:

```
Step Response Information:

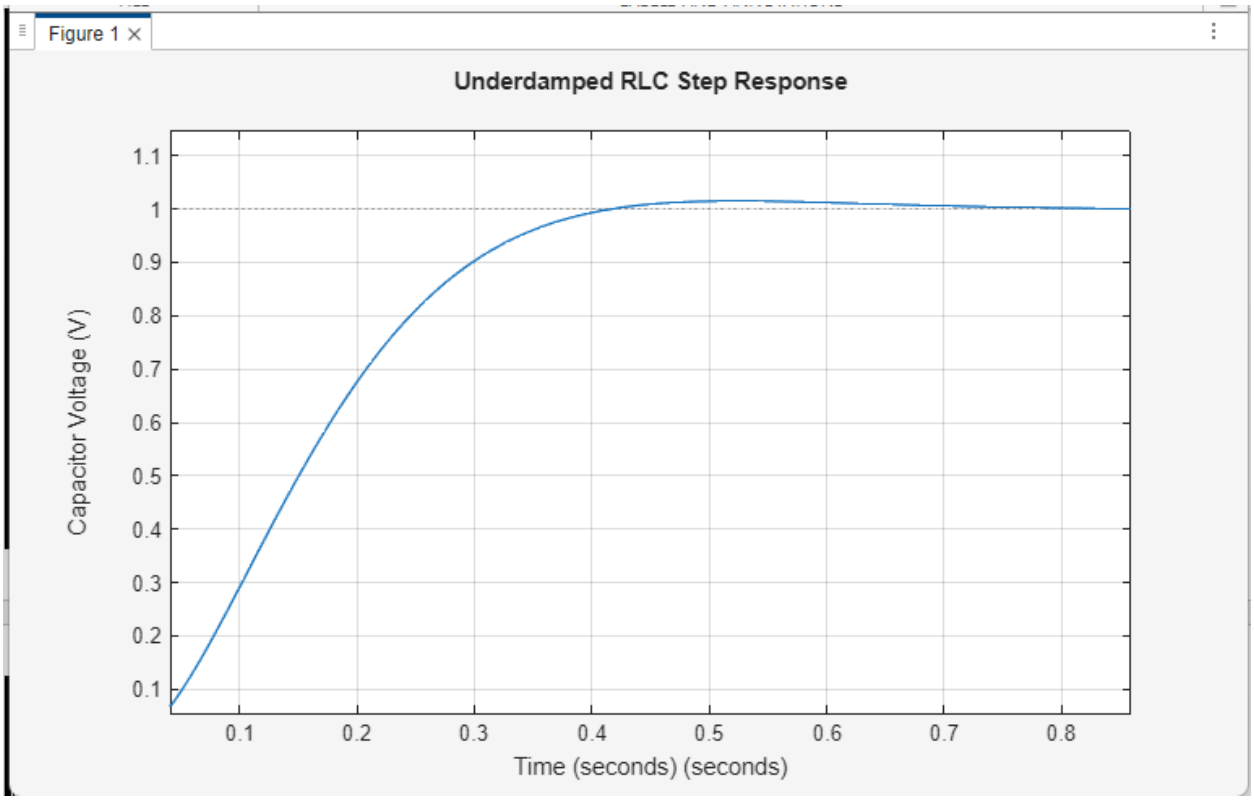
ans =

  struct with fields:

         RiseTime: 0.2468
    TransientTime: 0.3756
     SettlingTime: 0.3756
      SettlingMin: 0.9018
      SettlingMax: 1.0152
        Overshoot: 1.5165
       Undershoot: 0
             Peak: 1.0152
         PeakTime: 0.5238

>>
```

Figure:



Task#3:

Analyze a given RLC circuit Aseries RLC circuit has L = 0.8 H, C = 0.0125 F, and R = 4 Ω. Input is a unit-step voltage; output is capacitor voltage.□(□) = □_□(□)/□_□(□) = 1/(□□□² + □□□ + 1)

1. Derive the transfer function and calculate natural frequency and damping ratio.

2. UseMATLABtofind poles, damping class, DC gain, and stepinfo metrics.

3. Plot the step response and pole-zero map.

4. Verify final value and explain whether overshoot occurs.

5. Explain how R, L, and C appear in the denominator coefficients

Code:

```
1
2   clc;
3   clear;
4   close all;
5
6   R = 4;
7   L = 0.8;
8   C = 0.0125;
9
10  % Transfer function
11  G = tf(1, [L*C R*C 1]);
12
13  disp('Transfer Function:');
14  G
15
16  % System parameters
17  wn = 1/sqrt(L*C);
18  zeta = (R/2)*sqrt(C/L);
19
20  disp('Natural Frequency:');
21  disp(wn);
22
23  disp('Damping Ratio:');
24  disp(zeta);
25
26  disp('Poles:');
27  pole(G)
```

```
28
29      disp('Damping Verification:');
30      damp(G)
31
32      disp('DC Gain:');
33      dcgain(G)
34
35      disp('Step Response Metrics:');
36      stepinfo(G)
37
38      % Step response
39      figure;
40      step(G);
41      grid on;
42      title('RLC Circuit Step Response');
43      xlabel('Time (seconds)');
44      ylabel('Capacitor Voltage (V)');
45
46      % Pole-zero map
47      figure;
48      pzmap(G);
49      grid on;
50      title('RLC Circuit Pole-Zero Map');
51
```

Output:

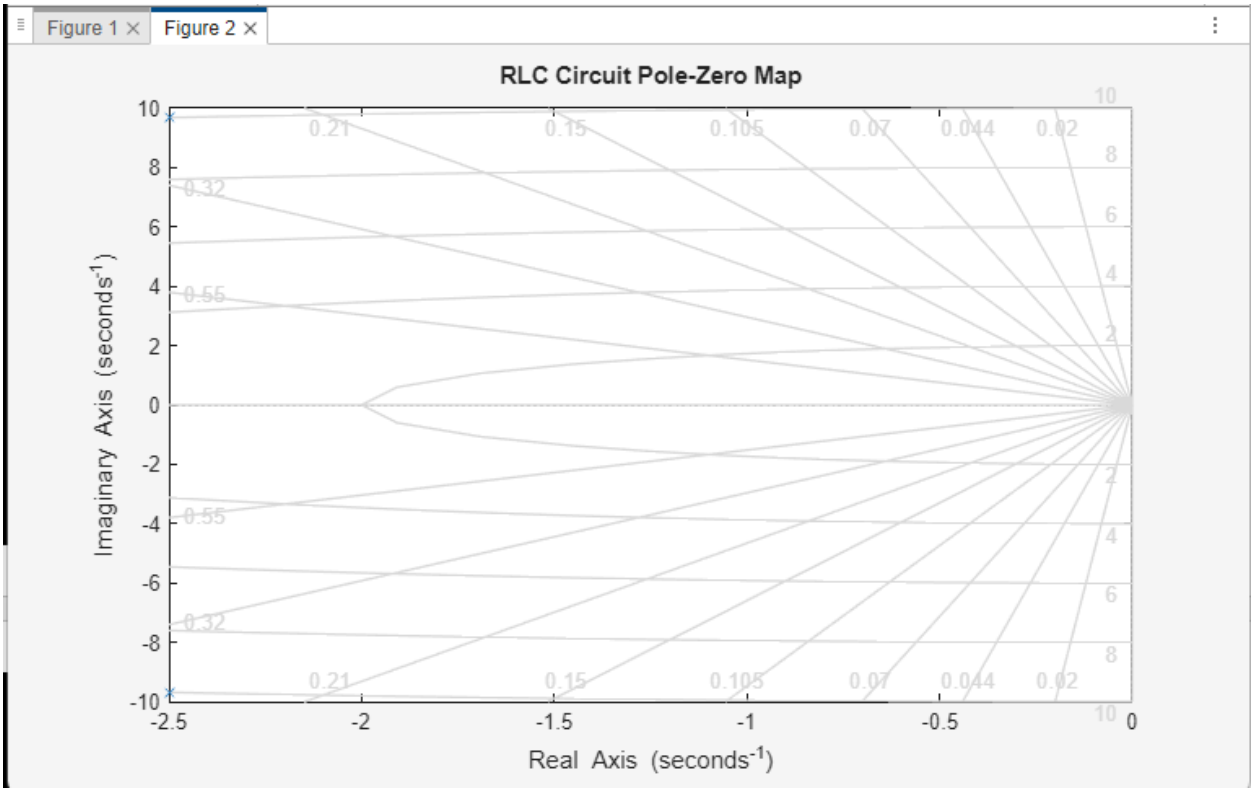
```
Step Response Metrics:

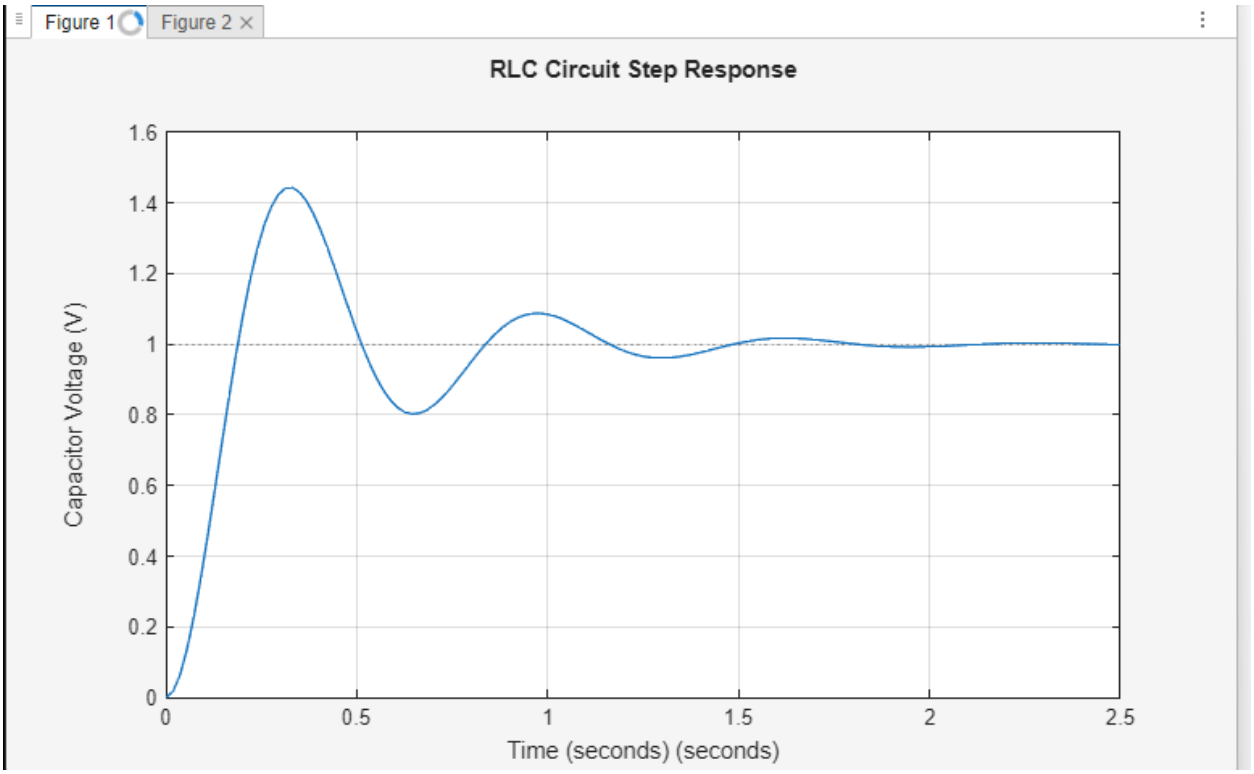
ans =

  struct with fields:

        RiseTime: 0.1269
    TransientTime: 1.4116
     SettlingTime: 1.4116
      SettlingMin: 0.8027
      SettlingMax: 1.4432
        Overshoot: 44.3235
       Undershoot: 0
             Peak: 1.4432
         PeakTime: 0.3316
```

Figure:


Figure 1
Figure 2
RLC Circuit Pole-Zero Map
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.21
0.15
0.105
0.07
0.044
0.02
0.32
0.55


Figure 1
Figure 2
RLC Circuit Step Response
Capacitor Voltage (V)
Time (seconds) (seconds)

Task #4:

Design a series RLC circuit with C = 0.01 F. Required unit-step overshoot is less than 10%, and 2%settling time is no more than 1.5 s.□□ < 0.10; □□ ≤ 1.5□.Use the standard approximation T_s ≈ 4/(ζω_n) for the 2% criterion.□_□ ≈ 4/(□□_□)

6. Choose a feasible damping ratio and natural frequency using second-order design relationships.

7. Calculate R and L. Control System- CP-402

8. Verify overshoot and settling time in MATLAB.

9. If the design fails either requirement, revise the component values and document the change.

10. Explain why satisfying the overshoot constraint does not guarantee the settling-time constraint

Code:

```
clc;
clear;
close all;

% Required capacitance
C = 0.01;

% Design specifications
zeta = 0.7;
wn = 4;

% Calculate circuit components
L = 1/(C*wn^2);
R = 2*zeta*sqrt(L/C);

% Construct transfer function
G = tf(1, [L*C R*C 1]);

disp('Designed Resistance R (Ohms):');
disp(R);

disp('Designed Inductance L (Henries):');
disp(L);

disp('Transfer Function:');
G

```

```
disp('Poles:');
pole(G)

disp('Damping Verification:');
damp(G)

disp('DC Gain:');
dcgain(G)

% Verify actual MATLAB response metrics
info = stepinfo(G);

disp('Overshoot (%):');
disp(info.Overshoot);

disp('Settling Time (seconds):');
disp(info.SettlingTime);

disp('Rise Time (seconds):');
disp(info.RiseTime);

% Check design requirements
if info.Overshoot < 10 && info.SettlingTime <= 1.5
    disp('Design satisfies both specifications.');
else
    disp('Design does not satisfy both specifications.');
end
```

```
figure;
step(G);
grid on;
title('Designed RLC Step Response');
xlabel('Time (seconds)');
ylabel('Capacitor Voltage (V)');

figure;
pzmap(G);
grid on;
title('Designed RLC Pole-Zero Map');
```

Output:

```
Designed Resistance R (Ohms):
    35

Designed Inductance L (Henries):
    6.2500

Transfer Function:

G =

             1
  ----------------------
  0.0625 s^2 + 0.35 s + 1

Continuous-time transfer function.
Model Properties
Poles:

ans =

  -2.8000 + 2.8566i
  -2.8000 - 2.8566i
```

```
Damping Verification:

         Pole              Damping       Frequency      Time Constant
                                       (rad/seconds)      (seconds)

 -2.80e+00 + 2.86e+00i     7.00e-01       4.00e+00         3.57e-01
 -2.80e+00 - 2.86e+00i     7.00e-01       4.00e+00         3.57e-01
DC Gain:

ans =

     1

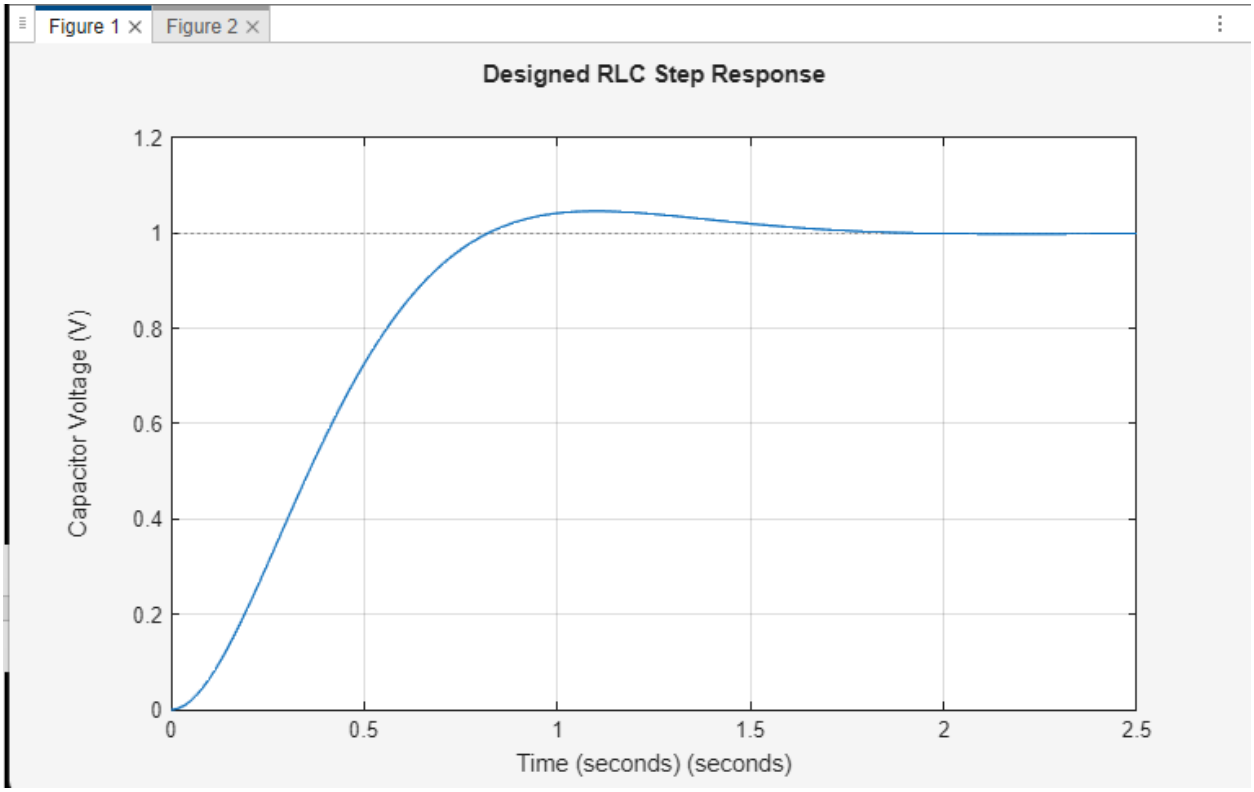
Overshoot (%):
    4.5986

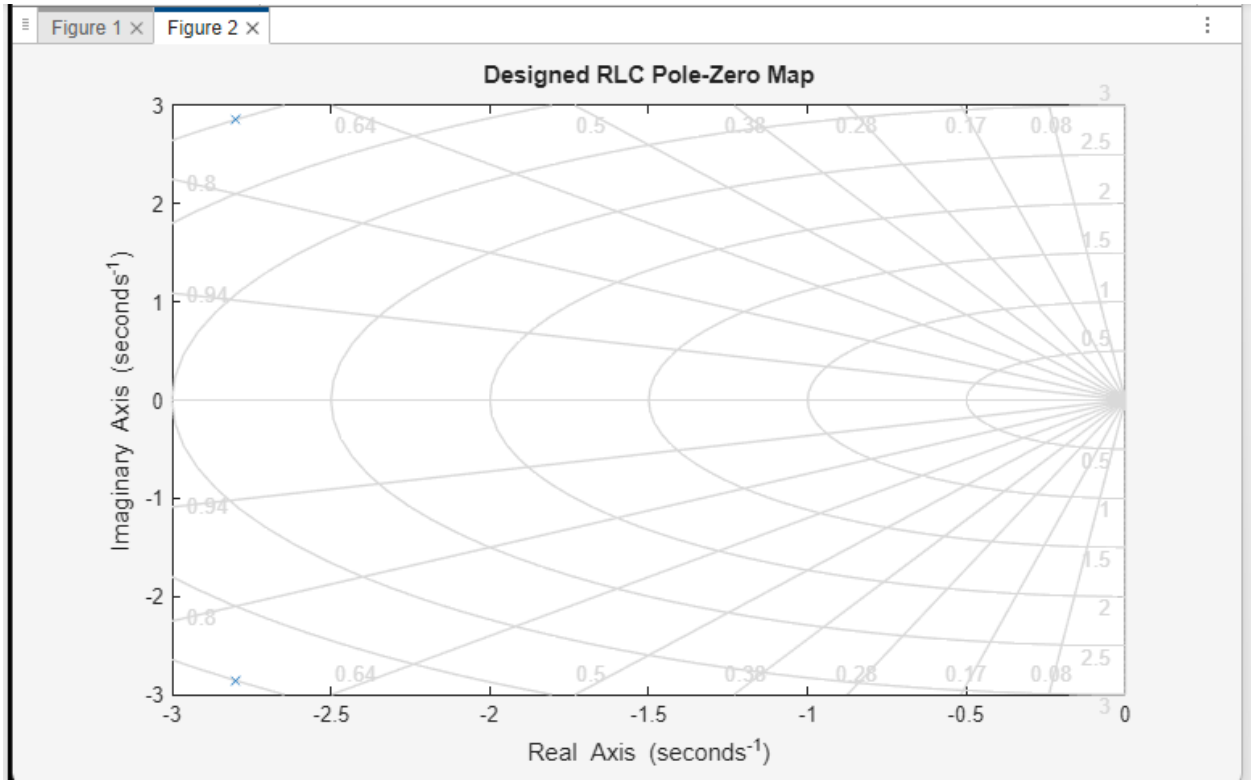
Settling Time (seconds):
    1.4947

Rise Time (seconds):
    0.5317

Design satisfies both specifications.
>>
```

Figure:


Figure 1
Figure 2
Designed RLC Step Response
Capacitor Voltage (V)
Time (seconds) (seconds)
0
0.2
0.4
0.6
0.8
1
1.2
0
0.5
1
1.5
2
2.5


Figure 1
Figure 2
Designed RLC Pole-Zero Map
Imaginary Axis (seconds-1)
Real Axis (seconds-1)
0.64
0.5
0.38
0.28
0.17
0.08
0.8
0.94
-3
-2.5
-2
-1.5
-1
-0.5
0
2.5
1.5
0.5